

WEEK 4 – ASSIGNMENT

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Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ Azure for Students

Resource group name * ⓘ cei-week4-Himanshu

Region * ⓘ (Asia Pacific) Central India

Previous Next Review + create

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Microsoft Azure

Search resources, services, and docs (G+)

Copilot

1000021453@dit.edu.in

Home > Resource Manager

Resource Manager | Resource groups

Summarize my costs by service

Export resource groups using Bicep or Terraform

How to manage changes with deployment tools?

Search

Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Add to service group

Group by none

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Filter for any field...

Subscription equals all

Location equals all

Add filter

☐	Name ↑	Subscription	Location
☐	cei-week4-Himanshu	Azure for Students	Central India

Showing 1 - 1 of 1. Display count: auto

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

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Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

The screenshot displays the Microsoft Azure portal interface. At the top, the header shows 'Microsoft Azure' with a search bar and user information. The breadcrumb trail indicates the path: Home > ceihbstorage01_1783836962294 | Overview. The main content area is titled 'ceihbstorage01' and 'Storage account'. A left-hand navigation pane lists various services, with 'Overview' selected. The main pane shows 'Essentials' with a table of account details:

Property	Value
Resource group	cei-week4-Himanshu
Location	centralindia
Subscription	Azure for Students
Subscription ID	12cfae79-a21e-4b1c-aaf6-2a55178e73a5
Disk state	Available
Performance	Standard
Replication	Locally redundant storage (LRS)
Account kind	StorageV2 (general purpose v2)
Provisioning state	Succeeded
Created	7/12/2026, 11:48:11 AM

Below the Essentials section, there are tabs for 'Properties', 'Monitoring', 'Capabilities (7)', 'Recommendations (0)', 'Tutorials', and 'Tools + SDKs'. The 'Properties' tab is active, showing two columns of settings:

Category	Setting	Status	
Blob service	Hierarchical namespace	Disabled	
	Default access tier	Hot	
	Blob anonymous access	Disabled	
	Blob soft delete	Enabled (7 days)	
	Container soft delete	Enabled (7 days)	
	Versioning	Disabled	
	Change feed	Disabled	
	NFS v3	Disabled	
	Allow cross-tenant replication	Disabled	
	Allow blob immutability	Disabled	
Security	Require secure transfer for REST API operations	Enabled	
	Storage account key access	Enabled	
	Minimum TLS version	Version 1.2	
Infrastructure encryption	Infrastructure encryption	Disabled	
	Networking	Public network access	Enabled
		Public network access scope	Enable from all networks
Private endpoint connections		0	

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Microsoft Azure

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A6D8DDDE-5798-4EBA-ASAB-78...

Home > ceihbstorage01_1783836962294 | Overview > ceihbstorage01

ceihbstorage01 Containers ☆ ...

Storage account

Search

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

> Security + networking

> Data management

> Settings

> Monitoring

+ Add container ↑ Upload ↻ Refresh 🗑️ Delete 🔒 Change access level ↺ Restore containers Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	📁 \$logs	7/12/2026, 11:48:34 AM	Private	Available	- ...
☐	📁 processed-data	7/12/2026, 11:51:47 AM	Private	Available	- ...
☐	📁 source-data	7/12/2026, 11:51:35 AM	Private	Available	- ...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

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A6D8DDDE-5798-4EBA-ASAB-78...

Home > ceihbstorage01_1783836962294 | Overview > ceihbstorage01 | Containers

source-data ...

Container

Search

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh 🗑️ Delete 📄 Copy 📄 Paste 🔄 Rename ⚙️ Acquire lease ⚙️ Break lease Edit columns

source-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	📄 Sample - Superstore.csv	7/12/2026, 11:52:42 AM	Hot (Inferred)	Block blob	2.18 MiB	Available ...

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Task 3: ADF Basics

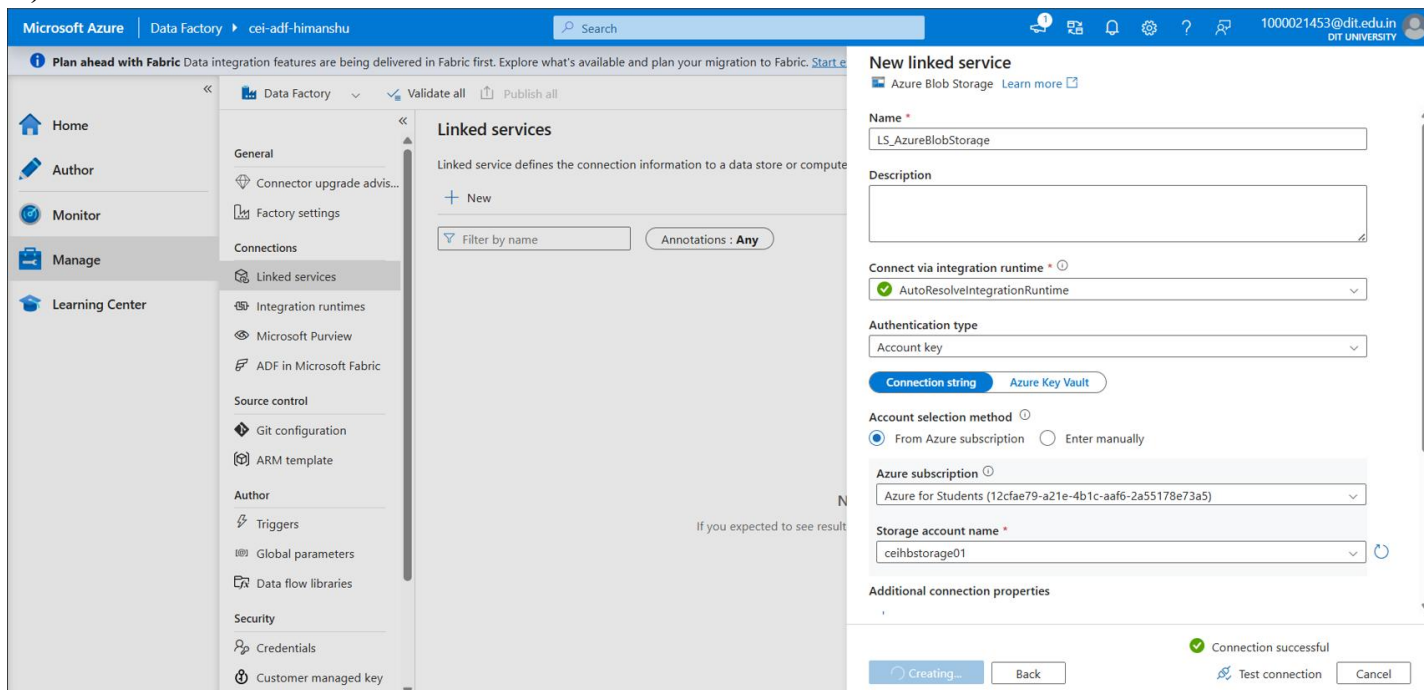
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

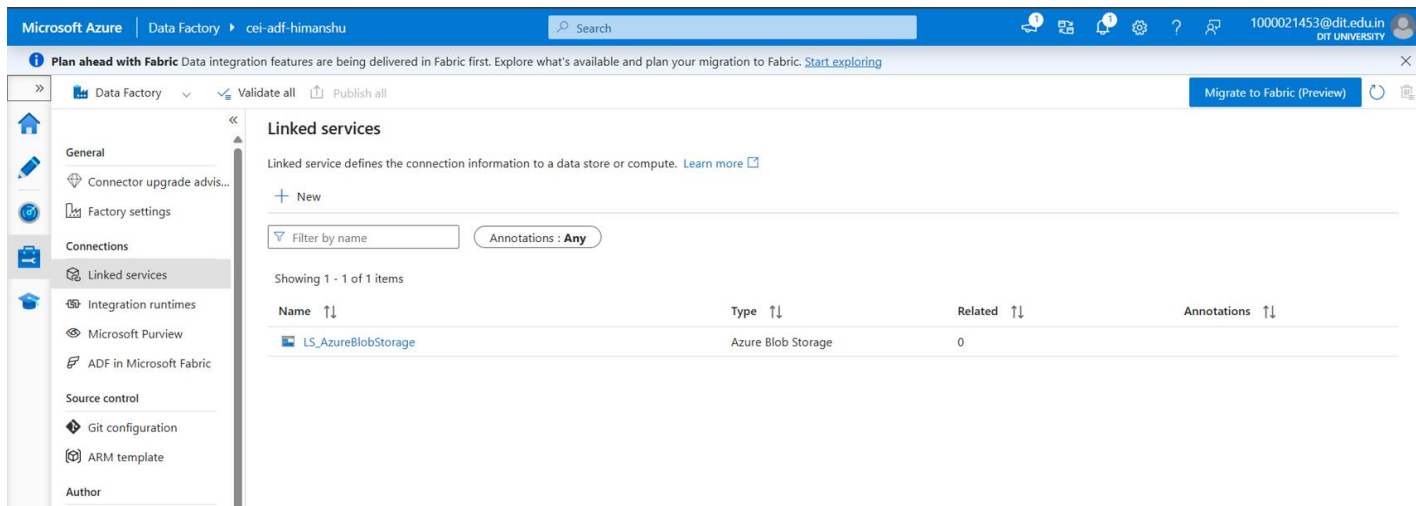
Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

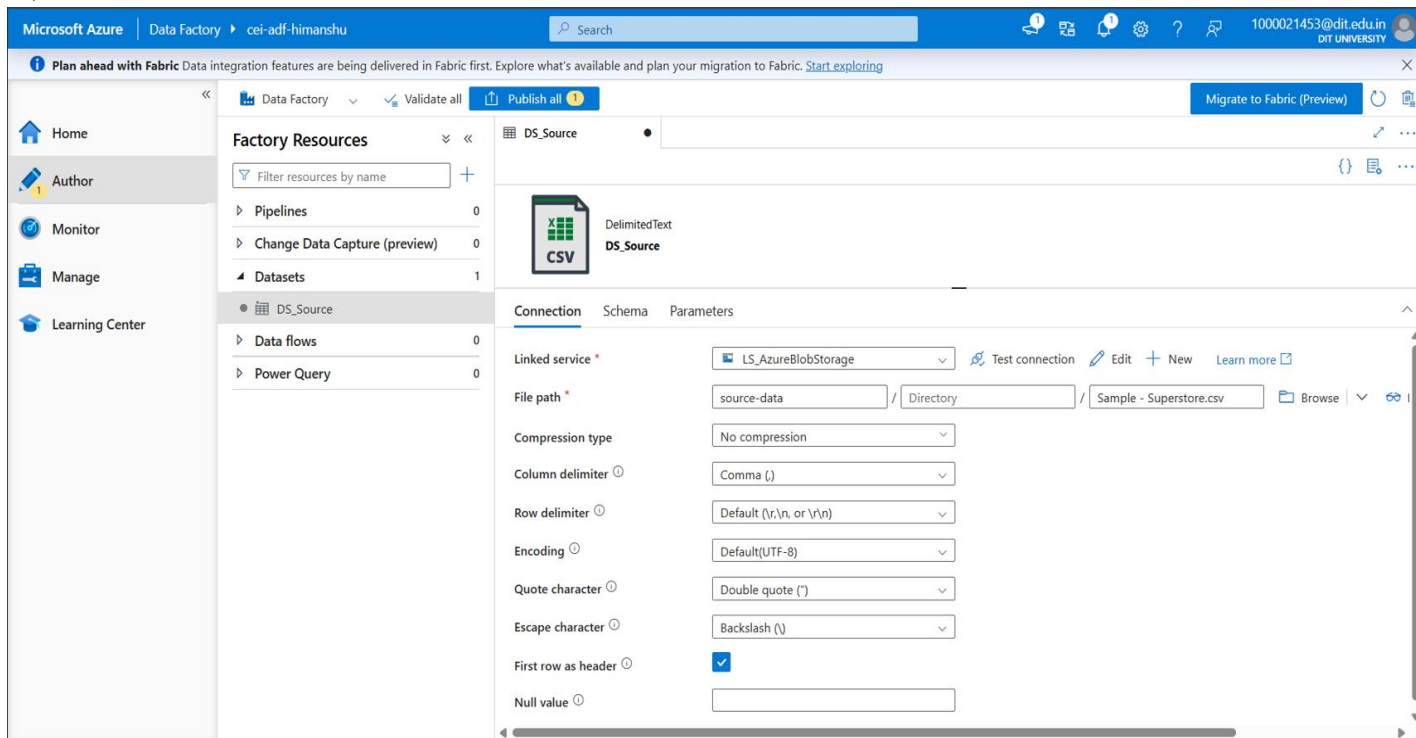
a.) Screenshot of Linked Service



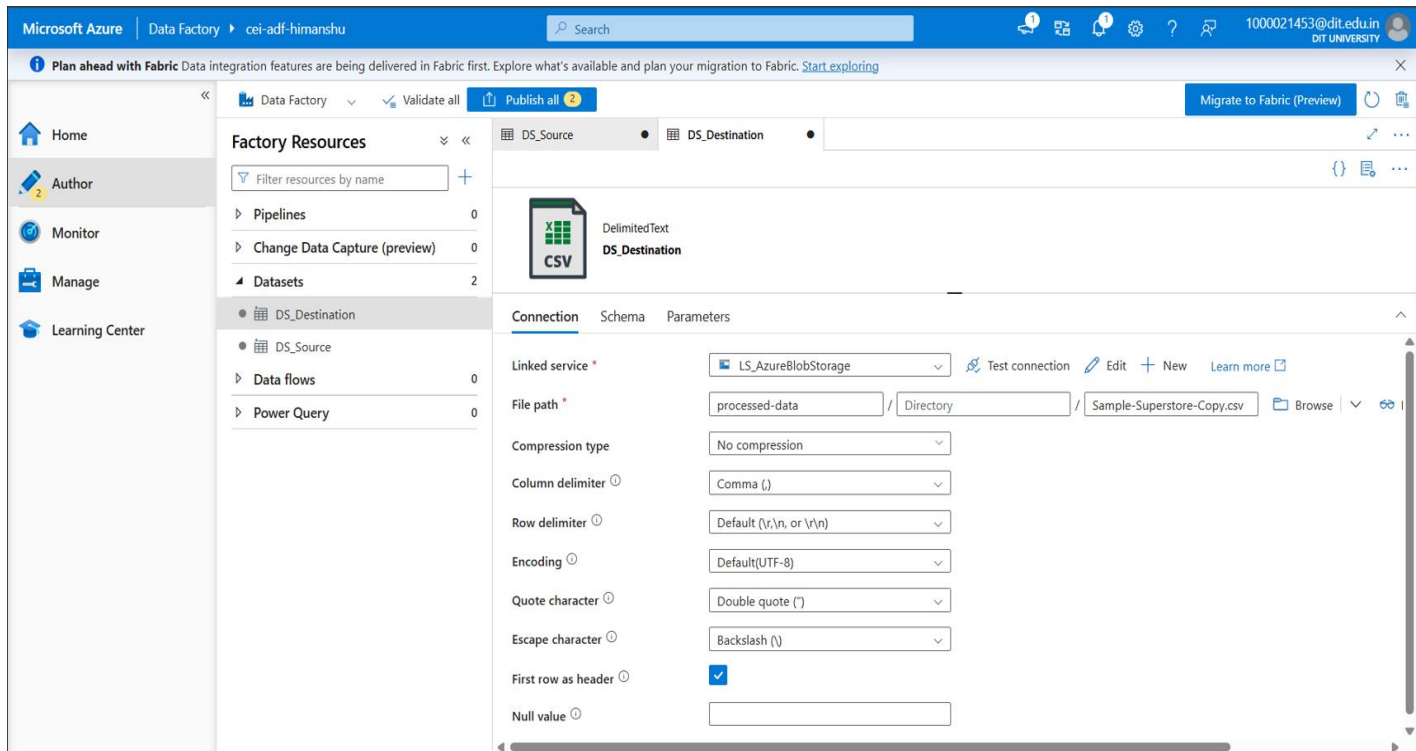
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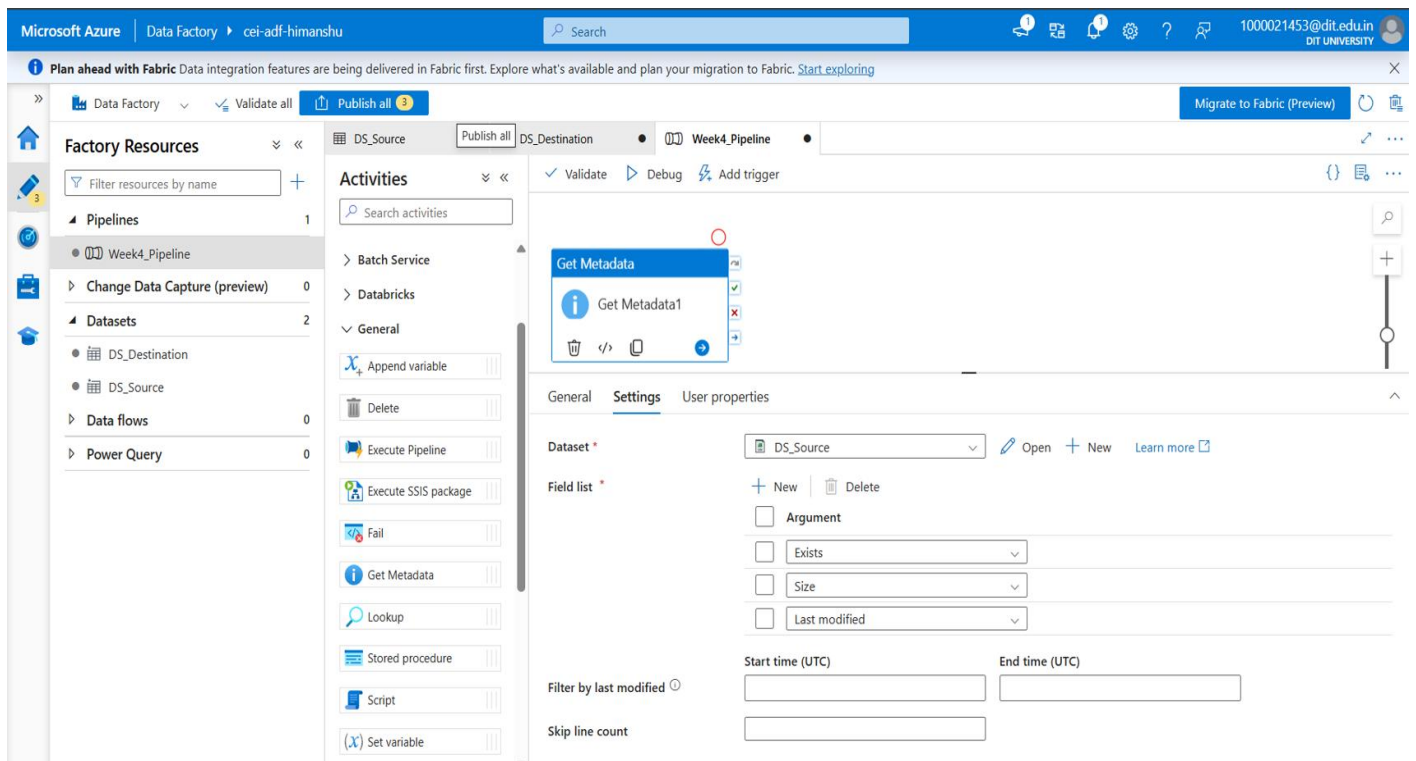
b.) Screenshot of DataSet



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c.) Screenshot of Get Metadata Activity



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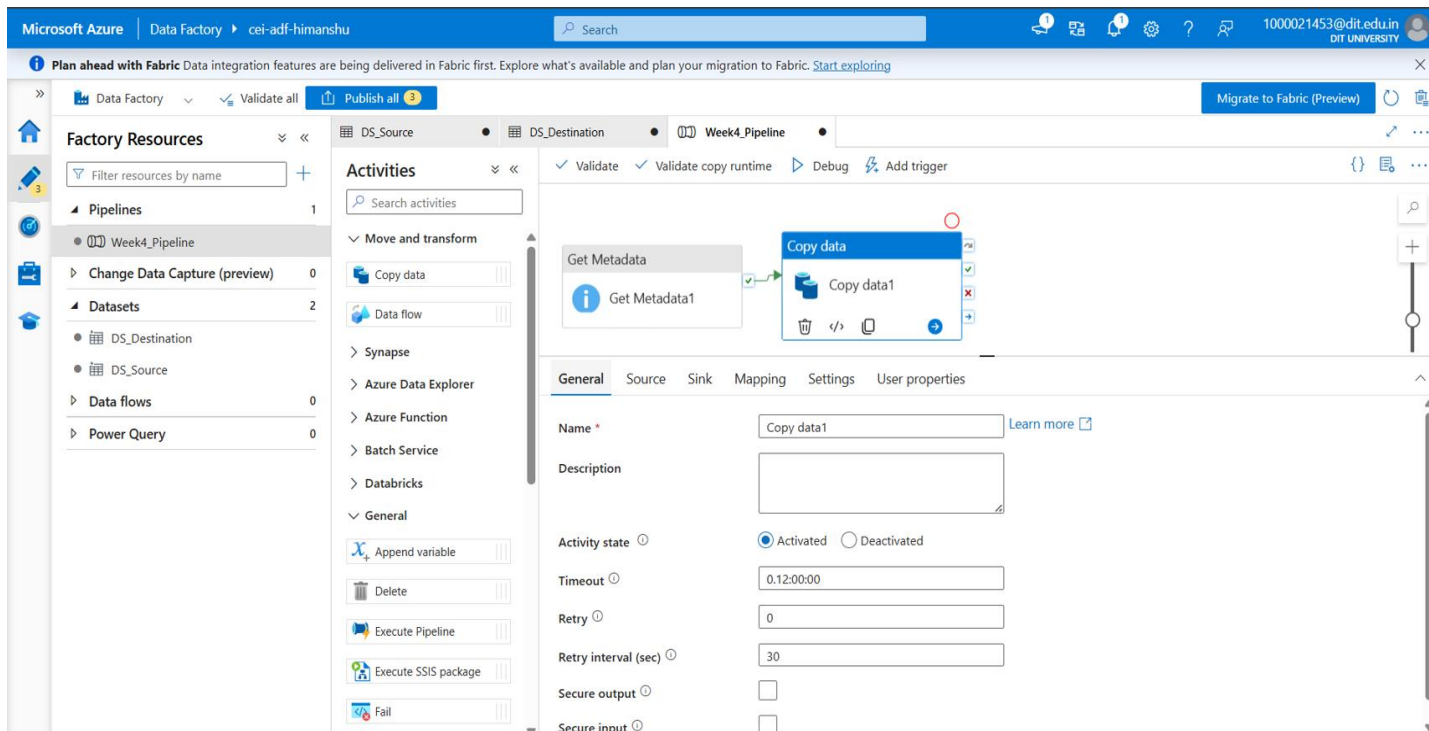
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



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This screenshot shows the Microsoft Azure Data Factory interface. The top navigation bar includes 'Microsoft Azure', 'Data Factory', and the user 'cel-adf-himanshu'. A search bar and a 'Migrate to Fabric (Preview)' button are also present. The left sidebar displays 'Factory Resources' with a tree view showing 'Pipelines' (1 item: 'Week4_Pipeline') and 'Datasets' (2 items: 'DS_Destination', 'DS_Source'). The main canvas shows a pipeline diagram with two activities: 'Get Metadata' (labeled 'Get Metadata1') and 'Copy data' (labeled 'Copy data1'). The 'Copy data' activity is selected, and its properties are shown in the right-hand pane. The 'Source' tab is active, displaying the following settings:

- Source dataset: DS_Source
- File path type: File path in dataset
- Start time (UTC):
- End time (UTC):
- Filter by last modified:
- Recursively: ☒
- Enable partitions discovery: ☐
- Max concurrent connections:
- Skip line count:
- Additional columns: + New

This screenshot shows the same Microsoft Azure Data Factory interface, but with the 'Sink' tab selected for the 'Copy data' activity. The 'Sink' tab displays the following settings:

- Sink dataset: DS_Destination
- Copy behavior: Preserve hierarchy
- Max concurrent connections:
- Block size (MB):
- Metadata: + New
- Quote all text: ☒
- File extension: .txt
- Max rows per file:

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Pipeline Validation

The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar shows 'Data Factory' and 'cei-adf-himanshu'. The left sidebar lists 'Factory Resources' including Pipelines (Week4_Pipeline), Change Data Capture (preview), Datasets (DS_Destination, DS_Source), Data flows, and Power Query. The main area shows the 'Week4_Pipeline' with two activities: 'Get Metadata' and 'Copy data'. The 'Copy data' activity is selected, and its 'General' tab is active, showing fields for Name, Description, Activity state (Activated), Timeout, and Retry. On the right, a 'Factory validation output' panel displays a green checkmark and the message: 'Your factory has been validated. No errors were found.' A 'Close' button is visible at the bottom right of this panel.

Pipeline Published

This screenshot shows the same Microsoft Azure Data Factory console as the previous one, but with a 'Publishing completed' notification. The notification box in the top right corner states: 'Publishing completed Successfully published'. The pipeline configuration and validation output panel remain the same, showing the 'Copy data' activity and the validation message: 'Your factory has been validated. No errors were found.' The 'Close' button is still present at the bottom right of the validation panel.

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Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'Week4_Pipeline'. The main area shows the 'Activities' tab for 'Week4_Pipeline', displaying a flow from 'Get Metadata1' to 'Copy data1'. Below the flow, the 'Output' tab shows the 'Pipeline run' details for ID 'f36306f3-1b71-4047-9558-f187ff12be48', indicating a 'Succeeded' status. A table below lists the activities:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	7/12/2026, 12:47:13 PM	16s	AutoResolveIntegrationRuntime (Central
Get Metadata1	Succeeded	Get Metadata	7/12/2026, 12:46:58 PM	14s	AutoResolveIntegrationRuntime (Central

The screenshot displays the 'Pipeline runs' section of the Microsoft Azure Data Factory console. The left sidebar shows the 'Monitor' tab selected. The main area shows the 'Pipeline runs' list for 'Week4_Pipeline'. The list includes columns for Pipeline name, Run start, Run end, Duration, Status, Triggered by, and Run ID. The first entry shows a 'Succeeded' status for the 'Week4_Pipeline' run.

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID
Week4_Pipeline	7/12/2026, 12:46:51	7/12/2026, 12:47:31	40s	Succeeded	Manual trigger	f36306f3-1b71-4047...

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Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

The screenshot shows the Microsoft Azure portal interface for the 'ceihbstorage01' storage account. The 'Access Control (IAM)' section is active, displaying the 'Role assignments' tab. The interface includes a search bar, filters for Type, Role, and Scope, and a table of role assignments. The table shows 4 results: 2 Owners (Himanshu Batra) and 2 Readers (cei-adf-himanshu).

Name	Type	Role	Scope	Condition
Owner (2)				
Himanshu Batra 1000021453@dit.edu.in	User	Owner	Subscription (Inherited)	None
Himanshu Batra 1000021453@dit.edu.in	User	Owner	Subscription (Inherited)	None
Contributor (1)				
cei-adf-himanshu e749c8c0-a560-430d-8c36-...	Managed identity	Contributor	Resource group (Inherited)	None
Reader (1)				
cei-adf-himanshu e749c8c0-a560-430d-8c36-...	Managed identity	Reader	This resource	None

Showing 1 - 4 of 4 results.

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Pipeline Execution Results

The Azure Data Factory pipeline was successfully validated, published, and executed using the **Debug** option. The **Get Metadata** activity completed successfully by validating the source CSV file's metadata, including its existence, size, and last modified timestamp. Following successful validation, the **Copy Data** activity transferred the CSV file from the `source-data` Blob container to the `processed-data` Blob container without any errors.

Execution Status

- **Pipeline Status:** ✓ Succeeded
- **Get Metadata Activity:** ✓ Succeeded
- **Copy Data Activity:** ✓ Succeeded
- **Metadata Validation:** ✓ Successful
- **Data Copy:** ✓ Successful
- **Output File:** Successfully created in the `processed-data` Blob container.

Observations

- The pipeline executed without any runtime errors.
- Source file metadata was successfully retrieved before data transfer.
- The destination file was generated successfully in Azure Blob Storage.
- End-to-end data movement between source and destination containers was completed successfully.

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Mini Project (Final Task)

Objective

Built an end-to-end Azure Data Factory pipeline to copy a CSV file from Azure Blob Storage while validating metadata before processing.

Implementation

- Created an Azure Storage Account and Blob Containers.
- Uploaded the Superstore CSV file to the source container.
- Configured Azure Blob Storage Linked Service.
- Created Source and Destination Datasets.
- Implemented a Get Metadata activity to validate the source file.
- Used Copy Data activity to copy the CSV into the destination container.
- Assigned Reader and Contributor IAM roles to the Azure Data Factory Managed Identity.
- Successfully executed and monitored the pipeline.

Result

- ✓ Source CSV successfully detected.
- ✓ Metadata validation completed successfully.
- ✓ CSV copied to the destination container.
- ✓ Pipeline executed without any errors.
- ✓ End-to-end Azure Data Factory pipeline completed successfully.

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a.) Screenshot of Pipeline Successfully Executed

The screenshot displays the Azure Data Factory (ADF) interface. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' expanded, listing 'Week4_Pipeline'. The main canvas shows the 'Week4_Pipeline' with two activities: 'Get Metadata1' and 'Copy data1', both marked with green checkmarks indicating success. Below the canvas, the 'Output' tab shows the 'Pipeline run' details for ID 'f36306f3-1b71-4047-9558-f187ff12be48', with a status of 'Succeeded'. A table below lists the activities and their execution details.

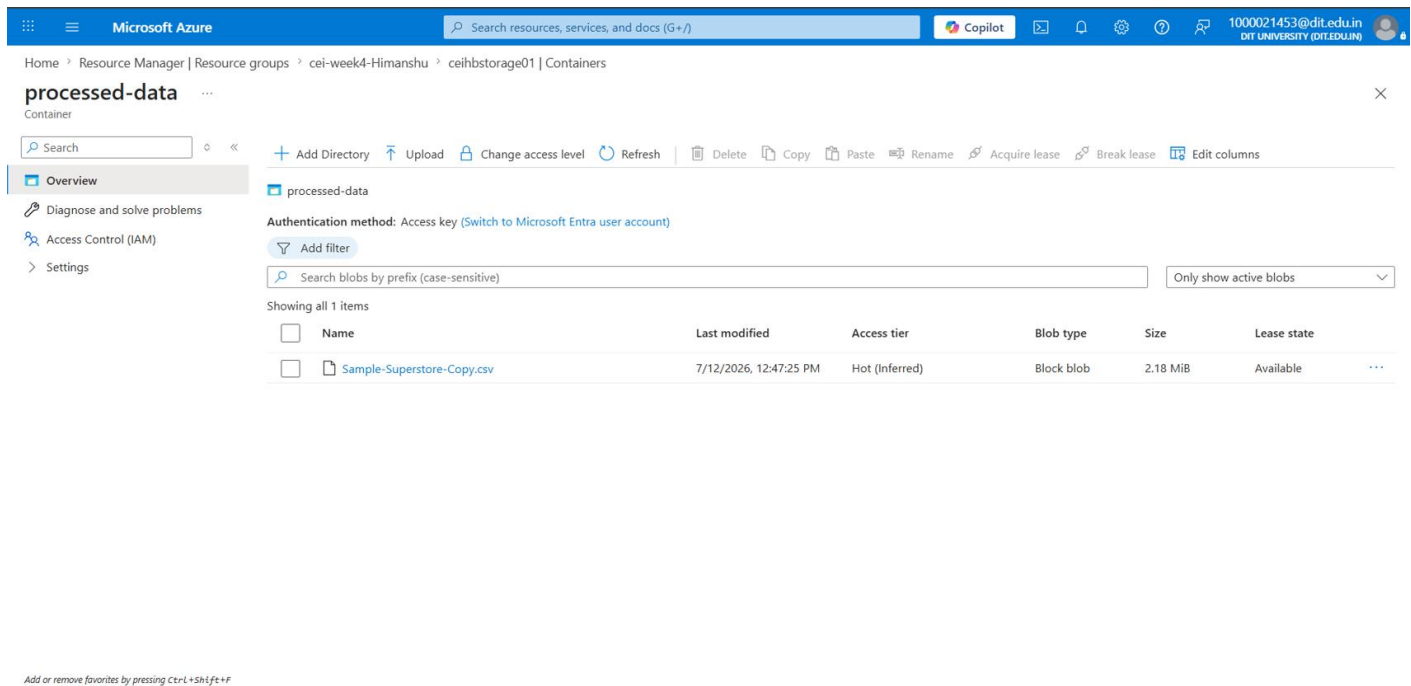
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/12/2026, 12:47:13 PM	16s	AutoResolveIntegrationRuntime (Central
Get Metadata1	✓ Succeeded	Get Metadata	7/12/2026, 12:46:58 PM	14s	AutoResolveIntegrationRuntime (Central

The screenshot shows the 'Pipeline runs' page in the Azure Data Factory interface. The left sidebar contains navigation options like 'Home', 'Author', 'Monitor', 'Manage', and 'Learning Center'. The main area displays a table of pipeline runs for 'Week4_Pipeline'. The table shows a single run that succeeded on 7/12/2026 at 12:47:34, triggered manually. The interface includes filters for 'Chennai, Kolkata, Mu...' and 'Last 24 hours', and options to 'Copy filters' and 'Export to CSV'.

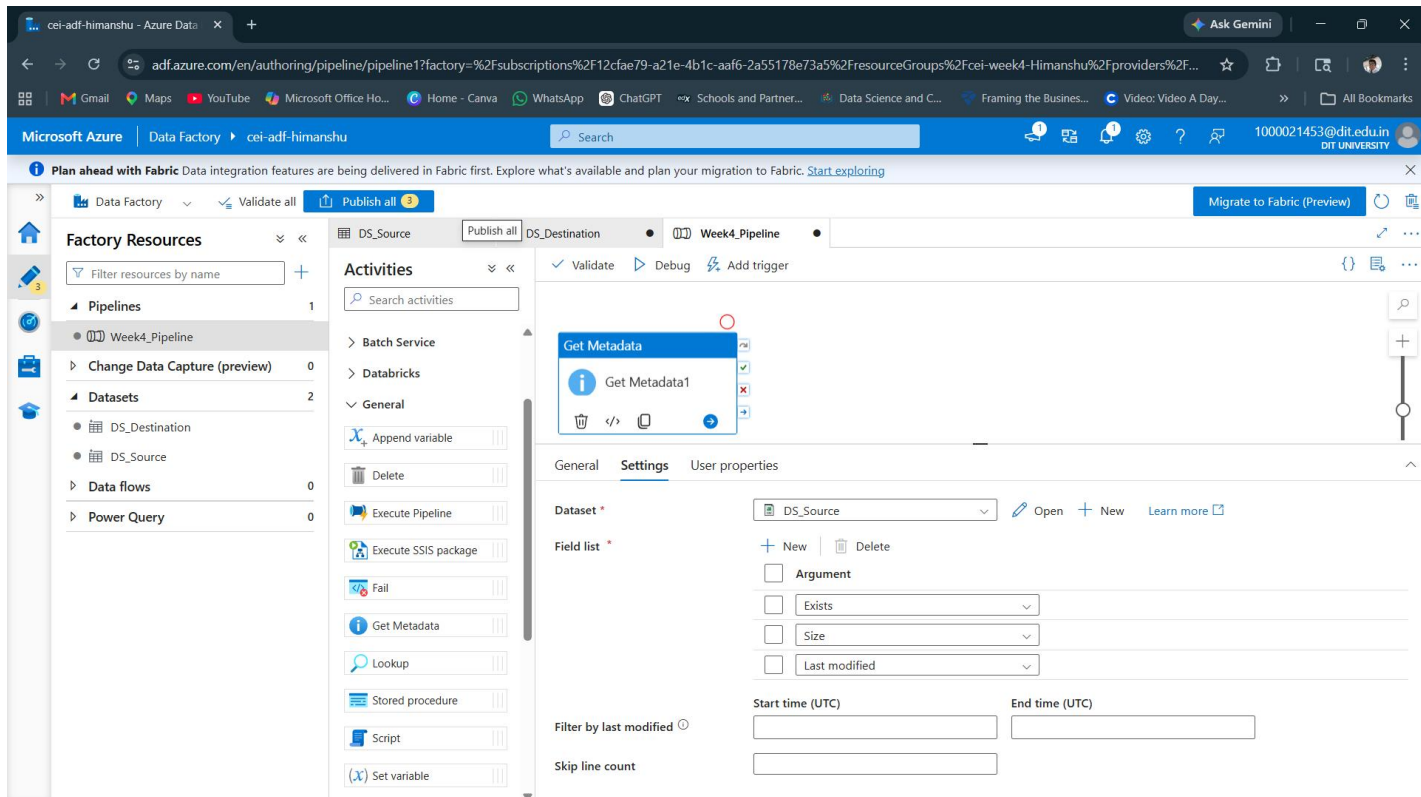
Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Param
Week4_Pipeline	7/12/2026, 12:46:54	7/12/2026, 12:47:34	40s	✓ Succeeded	Manual trigger	f36306f3-1b71-4047...	

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b.) Screenshot of Data Successfully Copied to Destination



c.) Screenshot of Get Metadata Activity



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d.) Screenshot of Metadata Successfully Validated

The screenshot displays the Azure Data Factory console. On the left, the 'Factory Resources' pane shows the hierarchy: Pipelines (1) > Week4_Pipeline. The main canvas shows the pipeline design with two activities: 'Get Metadata1' and 'Copy data1', both marked with green checkmarks. Below the canvas, the 'Output' tab is active, showing the pipeline run details. The pipeline run ID is f36306f3-1b71-4047-9558-f187ff12be48, and the status is 'Succeeded'. A table below lists the activities and their execution details.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	7/12/2026, 12:47:13 PM	16s	AutoResolveIntegrationRuntime (Central
Get Metadata1	Succeeded	Get Metadata	7/12/2026, 12:46:58 PM	14s	AutoResolveIntegrationRuntime (Central

Mini Project Summary

An end-to-end Azure Data Factory pipeline was developed to read a CSV file from Azure Blob Storage, validate its metadata using the **Get Metadata** activity, and copy it to a new destination container using the **Copy Data** activity. IAM roles (Reader and Contributor) were configured to enable secure access between Azure Data Factory and Azure Storage. The pipeline executed successfully, and the copied file was verified in the destination (processed-data) container, confirming successful data transfer and metadata validation.